

User Acceptance Testing (UAT) Template

Date	14 feb 2026
Team ID	LTVIP2026TMIDS35912
Project Name	Deep Learning Fundus Image Analysis for Early Detection of Diabetic Retinopathy
Maximum Marks	4 Marks

Project Overview

Project Name

Early Detection of Diabetic Retinopathy Using Deep Learning on Fundus Images

Project Description

Diabetic retinopathy is a severe eye disease caused by prolonged diabetes and is one of the leading causes of blindness worldwide. In its early stages, the disease often shows no visible symptoms, making timely diagnosis difficult. Traditional screening methods require expert ophthalmologists and specialized equipment, which are time-consuming and costly.

This project aims to improve early detection of diabetic retinopathy by applying deep learning techniques, specifically Convolutional Neural Networks (CNNs), on retinal fundus images. By analyzing high-resolution eye images and identifying hidden pathological patterns, the system accurately classifies disease stages, enabling early intervention, reducing vision loss, and supporting accessible and cost-effective healthcare solutions.

Project Version

DRVIP2025TMID35624

Testing Period

27-06-2025 to 28-06-2025

Testing Scope

Tested Functionalities

- **Prediction Endpoint (/api/predict)**
 - **Contact Form (/api/contact)**
 - **Image Upload & Validation**
 - **Navigation**
 - **Responsive UI**
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Requirements to be Tested

- **Postman – API testing**
 - **Web Browser – Navigation, form validation, UI testing**
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Testing Environment

URLs

- **Backend:**
<https://diabetic-retinopathy-backend.onrender.com/>
- **Frontend:**
<https://diabetic-retinopathy.netlify.app/>
- **Final Demo Link:**
<https://diabetic-retinopathy.netlify.app/>

Credentials

Not required

Test Cases

TC-001: Prediction Endpoint (/api/predict)

Test Steps

1. Open Postman
2. Select POST method
3. Enter URL:

`https://diabetic-retinopathy-backend.onrender.com/api/predict`

4. Go to Body → form-data
5. Upload a retinal fundus image
6. Click Send

Expected Result

```
{  
  "prediction": "Diabetic Retinopathy Detected",  
  "stage": "Moderate"  
}
```

Actual Result

```
{  
  "prediction": "Diabetic Retinopathy Detected",  
  "stage": "Moderate"  
}
```

Status:  Pass

TC-002: Contact Form (/api/contact)

Test Steps

1. Open Postman
2. Select POST method
3. Enter URL:

<https://diabetic-retinopathy-backend.onrender.com/api/contact>

4. Enter request body:

```
{  
  "name": "Garalaan Vishnuvardhan",  
  "email": "garalaanvishnu@gmail.com",  
  "subject": "Project Query",  
  "message": "Need clarification regarding diabetic retinopathy  
prediction"  
}
```

5. Click Send

Expected Result

```
{  
  "message": "Message sent successfully!"  
}
```

Actual Result

```
{  
  "message": "Message sent successfully!"  
}
```

Status:  **Pass**

TC-003: Image Upload & Form Validation

Test Steps

- **Open:** <https://diabetic-retinopathy.netlify.app/>

Expected Result

- **Only valid image formats allowed (JPG, PNG)**

- **Required fields must be filled**

Actual Result

- **Validation works as expected**

Status:  **Pass**

TC-004: Navigation

Test Steps

- **Open: <https://diabetic-retinopathy.netlify.app/>**

Expected Result

- **All navigation links (Home, About, Prediction, Contact) redirect properly**

Actual Result

- **All links working correctly**

Status:  **Pass**

TC-005: Responsive UI

Test Steps

- **Open application on desktop and mobile devices**

Expected Result

- **UI adapts properly across screen sizes**

Actual Result

- **Application is fully responsive**

Status:  **Pass**

Bug Tracking

Bug ID	Bug Description	Resolution	Severity	Status	Additional Feedback
BG-001	Low accuracy for minority DR classes	Applied data augmentation & class weighting	High	Closed	Improved multiclass performance
BG-002	Image upload delay	Optimized backend image preprocessing	Medium	Closed	Faster response time
BG-003	Model loading error in deployment	Used stable library versions	Low	Closed	Improved deployment stability

Sign-Off

Tester Name: Garalaan Vishnuvardhan

Date: 28-06-2025

Signature: Garalaan Vishnuvardhan